



# Experimental Investigation of PRANDTL-D3C Near-Wake

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The disappearance of the tip vortex in the near wake of a wing challenges conventional aerodynamic theories and presents new opportunities for drag reduction. Recent computational studies suggested that the PRANDTL-D3C wing—a swept, multi-element, and tapered wing with a bell-shaped lift distribution—exhibits this tip vortex disappearance. The current study explores the wake topology underlying this claim experimentally and numerically. Force-based and Particle Image Velocimetry (PIV) investigations were conducted on a 1:11 half-span scaled model of the PRANDTL-D3C wing to characterize the wake at various angles of attack. All experiments were performed in the University of Dayton Low Speed Wind Tunnel (UD-LSWT) using an open-jet test section configuration. Streamwise PIV was conducted at a chord-based Reynolds number of 185,000 across different spanwise stations to characterize the near-wake momentum deficit, and the non-planar wake formation. The experimental results showed a strong correlation between the wake velocity momentum deficit profiles and the FlightStream® panel method simulations, confirming reduced momentum deficit drag near the wingtips and higher aerodynamic efficiency compared to an elliptical lift distribution without a span increase at the design condition. Additionally, the analysis demonstrated that the near wake at this angle of attack lacks evidence of a trailing vortex, providing experimental validation of the computational predictions. However, to conclusively confirm the absence of the vortex, cross-stream PIV measurements are required to capture the full three-dimensional wake structure.

## Nomenclature

|           |                                  |                         |                                                  |
|-----------|----------------------------------|-------------------------|--------------------------------------------------|
| $\bar{c}$ | = Mean Aerodynamic Chord         | $\alpha_{\text{local}}$ | = Local angle of attack (deg)                    |
| $c$       | = Local chord                    | $\alpha_{\text{Wing}}$  | = Wing angle of attack based on root chord (deg) |
| $b$       | = Wingspan                       | $C_L$                   | = Coefficient of Lift                            |
| $U, V, W$ | = Velocity in X, Y, Z Directions | $C_D$                   | = Coefficient of Drag                            |

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## I. Introduction

The PRANDTL-D3C wing, based on Ludwig Prandtl's 1933 theory [1], can, under certain conditions enhance aircraft efficiency and facilitate coordinated turns without the need for rudder through its bell-shaped lift distribution. However, it is important to note that these advantages require a 22.5% increase in wingspan compared to wings with an elliptical lift distribution. For wings without an extended span (or with a span constraint), the elliptical lift distribution remains the optimal choice at the design  $C_L$ , which is mostly targeted at the cruise condition. In an elliptically loaded wing, the lift distribution gradient approaches infinity at the wingtip to satisfy zero circulation, resulting in strong tip vortex formation [2]. A marked discontinuity exists between the upwash and downwash at the wingtip, indicative of a strong vortex. However, with the bell-shaped lift distribution, due to the shift of the center of lift inboard of the wing, the wingtips are lightly loaded, which can result in a near zero-lift distribution gradient at the wingtips. This is evidenced by the smooth transition between the upwash and downwash, occurring inboard of the wing and producing a "wing vortex" rather than a wingtip vortex. The presence of a wing vortex has been confirmed through flight tests [3] and wind tunnel tests [4] across a range of operating conditions.

Earlier experimental investigations have shown that the rollup of wingtip vortices and their separation from the shear layer directly impact a wing's aerodynamic efficiency [5] [6]. Various devices, such as winglets and both active and passive wingtip devices, have been used to mitigate the rollup and growth of these vortices. However, the Richardson number at the core of the vortices suggests that turbulence induced by these devices is suppressed and has minimal effect on the rollup process [7]. Despite this, some studies have demonstrated complete vortex mitigation in the wake of wings with a non-elliptical lift distribution. Ranjan and Ansell [8] conducted a RANS study comparing the wakes of wings with elliptical and bell-shaped lift distributions, each producing the same total lift. Significant differences were observed in the wake. The elliptical wing produced a planar wake with a pair of strong trailing vortices, following the typical pattern of wake rollup, as shown in Figure 1a. In contrast, the wing with the bell-shaped lift distribution exhibited a non-planar wake with a slight "dihedral" angle—a feature solely due to induced velocity, as the wing used to generate these results has no dihedral angle. Furthermore, no rollup of vortices was observed in the near wake, as depicted in Figure 1b. A comprehensive far-field study by the authors indicated that vortices eventually rolled up, but only at a downstream distance of  $100 x/c$ , compared to  $15 x/c$  for the elliptically loaded wing.

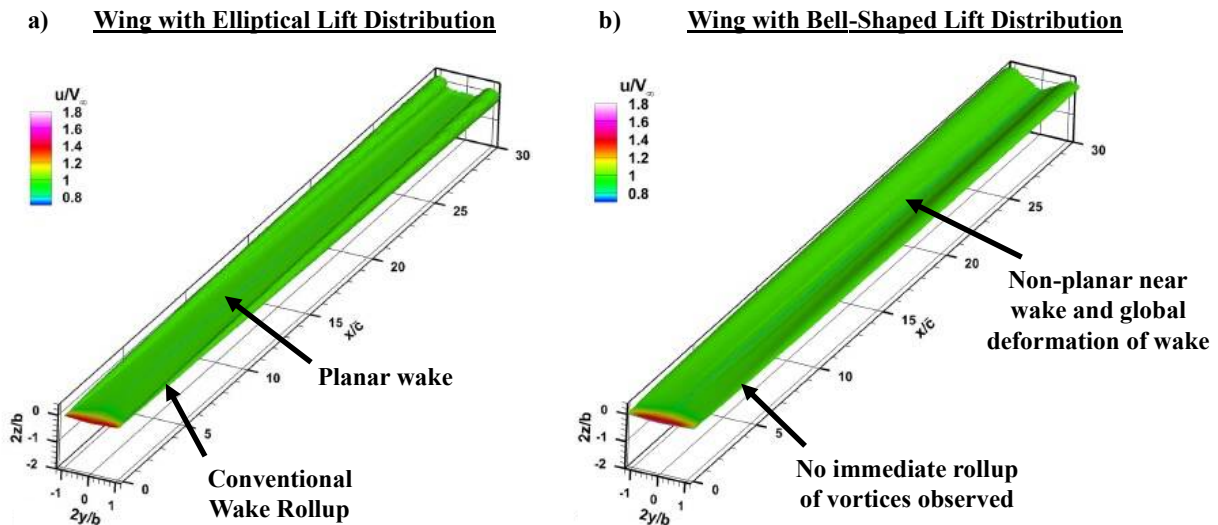


Figure 1 Wake roll-up process for a wing with a) Elliptical lift distribution b) Bell-shaped lift distribution [8]

A similar RANS study was conducted by Hammer and Garmann [9] in the near field of a wing with a bell-shaped lift distribution, using a wing geometry similar to that of the PRANDTL-D3C aircraft [3]. The simulations were performed at two Reynolds numbers: 343,000 and 3,000,000, with angles of attack ranging from  $-4^\circ$  to  $16^\circ$ . The near-wake and on-body velocity results are shown in Figure 2. At an angle of attack of  $8^\circ$ , no trailing vortices were observed at either Reynolds number. At this angle of attack, the wing's aerodynamic efficiency reached its maximum. For other angles

of attack, contrary to the results shown in [3], the trailing vortex did not shift inboard to form a wing vortex but remained at the wingtip. The theoretical relationship between the non-elliptical lift distribution and the wake characteristics is described in a recent study by Hoeijmakers et al. [10] where the authors inverted the Biot-Savart principle to compute circulation distributions from specified upwash and downwash profiles. This relationship provides insights into singularities in tip-region upwash and wake vortex sheet roll-up in an idealized scenario such as thin wing and neglect of three-dimensional interactions. Most of the wake signature analysis of the PRANDTL-D3C wing has been conducted through computational investigations, and to the best of the authors' knowledge, no experimental campaign has yet been carried out to characterize the wake of the PRANDTL-D3C wing. Therefore, the experimental objective of the current study is to quantitatively characterize the wake of the PRANDTL-D3C wing at different angles of attack and to compare these results with the RANS simulation results conducted by Hammer and Garmann [9].

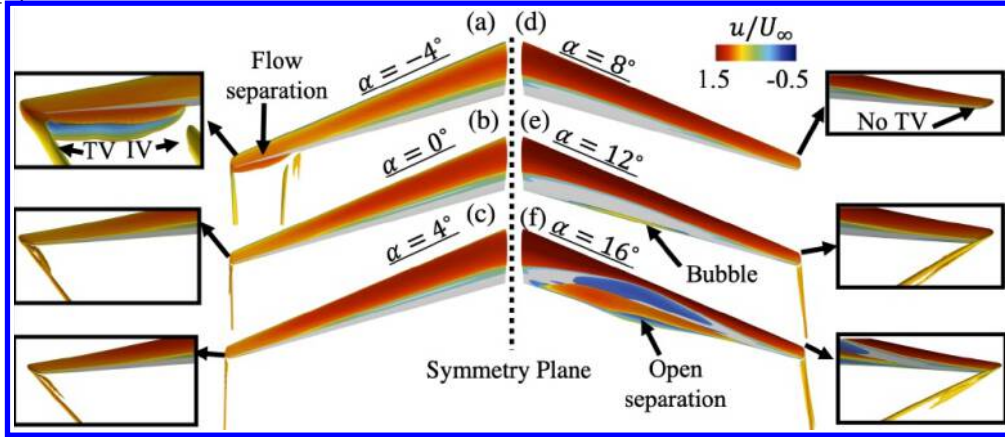


Figure 2 Isosurfaces of Q-criterion of U velocity showing tip vortex (TV) and inboard vortex (IV) at different angles of attack [9]

## II. Experimental setup

### A. University of Dayton Low Speed Wind Tunnel

All experiments were conducted in the University of Dayton's Low Speed Wind Tunnel (UD-LSWT) using its open-jet test section configuration. The tunnel is an open return type and has a 16:1 contraction ratio, six anti-turbulence screens, and a 0.762 m by 0.762 m open-jet test section. The effective length of the test section is 0.183 m. A 1.128 m by 1.128 m collector gathers the expanded air into the diffuser. A pitot tube is connected to a Siemens QBM-3230-U5D analog pressure transducer to measure the freestream velocity during the experiments. A TSI T600 Micromanometer was used to calibrate the pressure transducer, and an NI-USB-6259 DAQ was used to collect the pressure transducer data at the same sampling rate as the F/T transducer.

### B. Force-based Testing

A 1:11 scaled model of the PRANDTL-D3C wing was fabricated using 3D printing. The wing's dimensions and twist distribution were obtained from [9] and [3]. The model has a 25-inch span and a mean aerodynamic chord of 3.5 inches and was tested at a Reynolds number of 185,000. Images of the experimental setup are shown in Figure 3. The wing model was mounted to a rotary stage and an ATI Gamma F/T balance assembly. The sensor's calibrated uncertainty was 1.25% for the Y-axis, which measures the wing's normal force, and 0.75% for the X-axis, which measures the wing's axial force. A 0.914 m x 1.829 m splitter plate enabled half-span model testing of the wing.

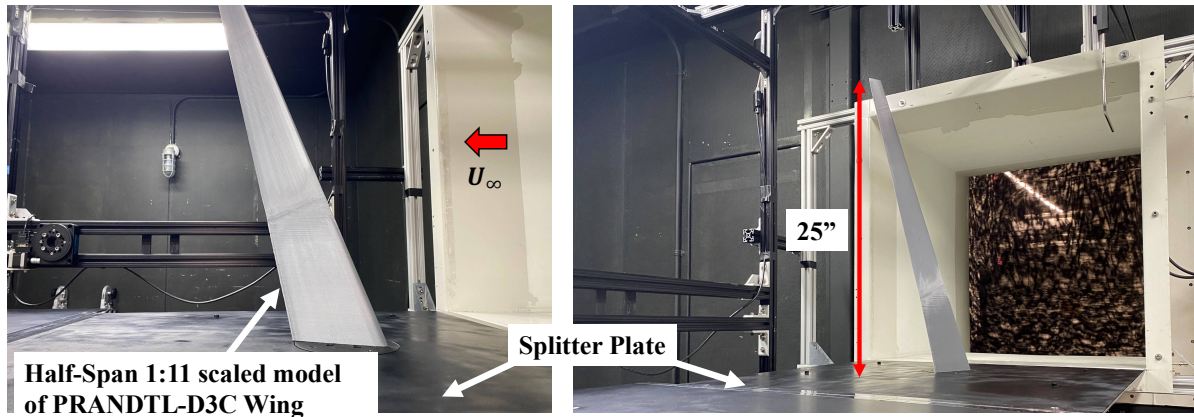


Figure 3 Pictures of force-based experimental setup of 1:11 scaled model of PRANDTL-D3C wing at the University of Dayton Low Speed Wind Tunnel

### C. PIV Setup

Streamwise Particle Image Velocimetry (PIV) was conducted downstream of the trailing edge to capture the wake at various locations along the span, as shown in Figure 4a. Due to the sweep angle of the model the captured field of view (FOV) is located at different downstream distances from the wing trailing edge. The distance from the trailing edge is furthest at the root showing a more developed wake. The experiment utilized a ViCount smoke machine to seed the wind tunnel with particles. A Quantel ND-YAG dual-pulse laser (Quantel Twin CFR 300), in combination with a plano-concave lens and a mirror, was used to produce a laser sheet that illuminated a streamwise plane. The laser and an Imperx B2021 camera with a 100 mm lens were triggered simultaneously using a Quantum Composer pulse generator. The PIV setup in the UD-LSWT is shown in Figure 4b. The time delay between laser pulses was set to achieve an 8 to 10-pixel particle displacement in the free shear layer. For each run, 1,000 image pairs were captured and processed using Fluere [11]. Two iterations were performed in the DPIV processing: the first iteration used 64-pixel interrogation windows, and the second iteration used 32-pixel interrogation windows. All components were mounted on an optical bench supported by a linear rail to allow for spanwise translation.

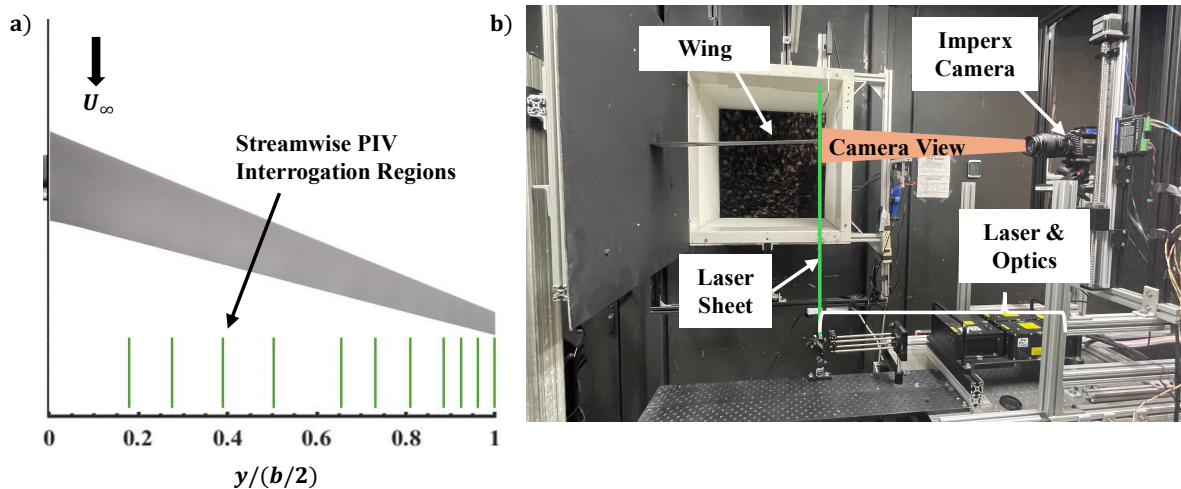


Figure 4 a) Local Y location of PIV planes along the spanwise direction b) PIV setup in the UD-LSWT

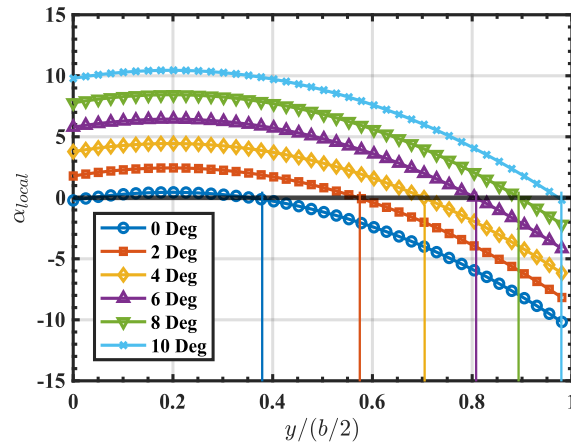
The test matrix for the spanwise PIV is shown in Table 1. Eleven different spanwise stations were interrogated, with refined spacings between the interrogation planes closer to the wingtip. The field of view (FOV) was consistent for the first eight regions. Since the wing model has a  $5^\circ$  dihedral, the FOV for the spanwise stations closer to the root



had to be lowered in the Z-direction and was made slightly larger than the outboard FOV. The Reynolds number was maintained at 185,000 for all cases, and PIV was conducted at each spanwise station at six different angles of attack. Note that  $\alpha_{wing}$  in Table 1 is defined as the angle of attack at the root chord of the wing, consistent with the nomenclature used by Hammer and Garmann [9].

**Table 1. Test Matrix for PIV Experimental Investigations**

| Spanwise Distance from Wingtip (in) | y/(b/2) | Freestream (m/s) | Reynolds Number | FOV (mm)  | $\alpha_{wing}$ (Deg) |
|-------------------------------------|---------|------------------|-----------------|-----------|-----------------------|
| 0                                   | 1.00    | 15               | 185,000         | 101 x 101 | 0, 2, 4, 6, 8, 10     |
| 1                                   | 0.96    |                  |                 |           |                       |
| 2                                   | 0.92    |                  |                 |           |                       |
| 3                                   | 0.88    |                  |                 |           |                       |
| 5                                   | 0.80    |                  |                 |           |                       |
| 7                                   | 0.72    |                  |                 |           |                       |
| 9                                   | 0.64    |                  |                 |           |                       |
| 13                                  | 0.48    |                  |                 |           |                       |
| 16                                  | 0.36    |                  |                 | 109 x 109 |                       |
| 19                                  | 0.24    |                  |                 |           |                       |
| 21.5                                | 0.18    |                  |                 |           |                       |



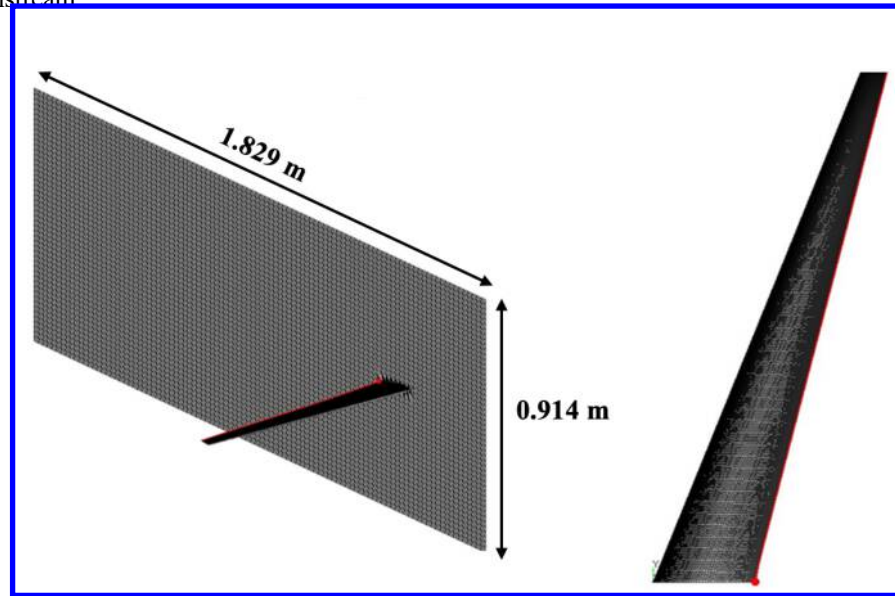
**Figure 5 Variation of local angle of attack along the wingspan**

Figure 5 shows how  $\alpha_{local}$  decreases toward the wingtip. The twist is nonlinear by design in order to achieve the PRANDTL-D3C wing's bell-shaped lift distribution, where lift is concentrated inboard. This nonlinear twist distribution results in reduced local angles of attack near the wingtip, potentially influencing the strength and positioning of wake vortices. As  $\alpha_{wing}$  increases, the spanwise regions with zero or negative  $\alpha_{local}$  extend further outboard, suggesting a shift in the wake characteristics. These spanwise variations are expected to correlate with the wake behavior discussed in later sections.

#### D. Numerical Investigation with FlightStream®

FlightStream® [11] was used to numerically investigate the same model geometries tested in the wind tunnel to obtain the surface pressure distribution on the PRANDTL-D3C wing. FlightStream® includes two main types of potential flow solvers: a pressure-based solver and a vorticity solver. The pressure-based solver, which utilizes the pressure

fields on the model geometry to determine the aerodynamic loads, is used in the present study. A combination of source and sink panels, along with doublet panels on the trailing edge, is employed to provide a solution for the pressure-based potential flow solver. For the current study, the pressure-based solver was selected based on its good agreement with the wind tunnel experimental results in our previous studies [12–16]. The boundary layer is assumed to be fully turbulent, with viscous coupling enabled. The surface mesh of the wing and the splitter plate in FlightStream® is shown in Figure 6. The wing platform has a structured mesh with 120 node points spanwise and 80 node points chordwise on both the upper and lower surfaces. The chordwise mesh also has a dual-side growth rate of 1.05 to better resolve the leading and trailing edge geometry. Additionally, a splitter plate of the same size as the one used in the wind tunnel experiment is included in the model to better represent the actual experimental conditions (minus the boundary layer growth and interaction). The splitter plate has 60 node points from side to side and 120 node points downstream



**Figure 6 Surface Mesh in FlightStream® for a) Isometric View of the Wing and Splitter Plate and b) Top View of the Wing**

### III. Results

#### A. Force Based Results

The aerodynamic coefficients comparing results from experiments, FlightStream® simulations, and findings from Hammer and Garmann [9] are shown in Figure 7. The  $C_L$  results show good agreement between the simulations up to the point of stall. Differences in the stall angle of attack may be attributed to variations in Reynolds numbers between the simulations conducted by Hammer and Garmann ( $Re = 300,000$ ) and the experiments performed at the UD-LSWT ( $Re = 185,000$ ). The drag coefficients from the various methods also show good agreement, with the highest drag observed in the experiments, as expected. Since the FlightStream® results closely align with the experimental data, they can be utilized to visualize the wake rollup process.

The near-wake simulation results are compared between Hammer and Garmann [9] and FlightStream® at an  $8^\circ$  angle of attack, as shown in Figure 8. Both simulations indicate that no trailing vortices are present at this angle of attack, which appears to be a unique feature of the PRANDTL-D3C wing's bell-shaped lift distribution. The absence of trailing vortices suggests reduced drag. The mitigation of strong wingtip vortices observed in elliptically loaded wings can be attributed to shifting the lift distribution inboard and reducing the loading at the wingtips. Furthermore, as illustrated in Figure 8c, the  $8^\circ$  angle of attack corresponds to the point of maximum aerodynamic efficiency ( $L/D$ ), as evidenced by all three investigation methods (experiment, FlightStream®, and RANS). The variations in the maximum  $L/D$  between different cases are attributable to slight differences in predicting  $C_D$ , as shown in Figure 7b.

Nevertheless, the correlation between the angle of attack with no trailing vortices and maximum L/D provides insight into the optimal aerodynamic efficiency of the PRANDTL-D3C design. However, the factors underlying the disappearance of the wingtip vortex at this angle of attack remain experimentally unproven and the associated aerodynamic features leading to this condition are yet undescribed in the literature. Addressing this gap is one of the research objectives of the current study.

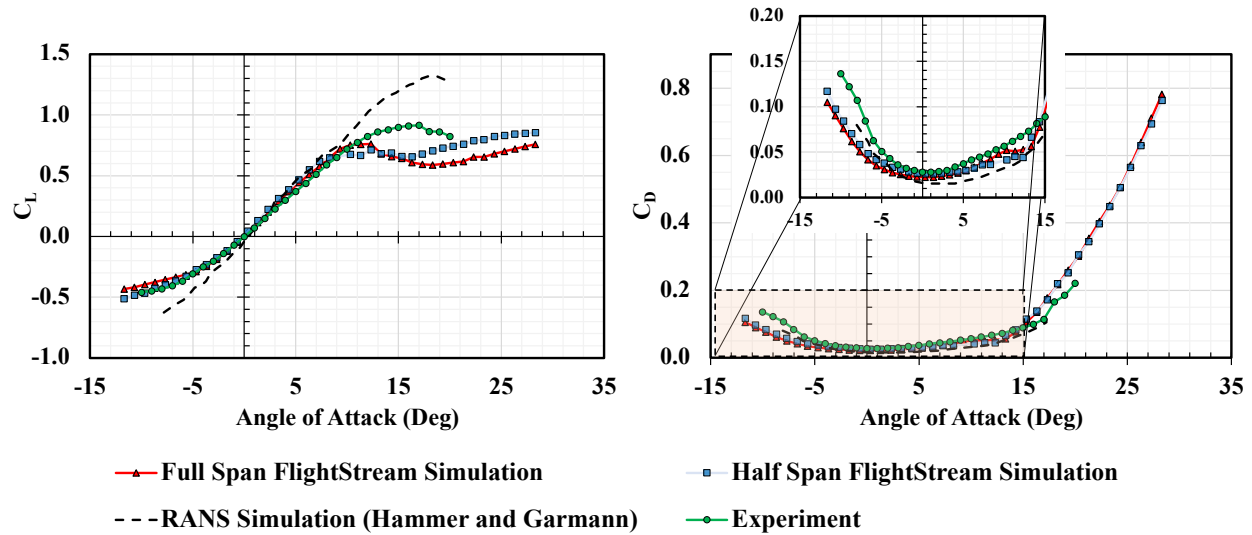


Figure 7 Comparisons of Coefficient of lift and coefficient of drag as a function of angle of attack among experiments, FlightStream® simulations and RANS simulation from [9]

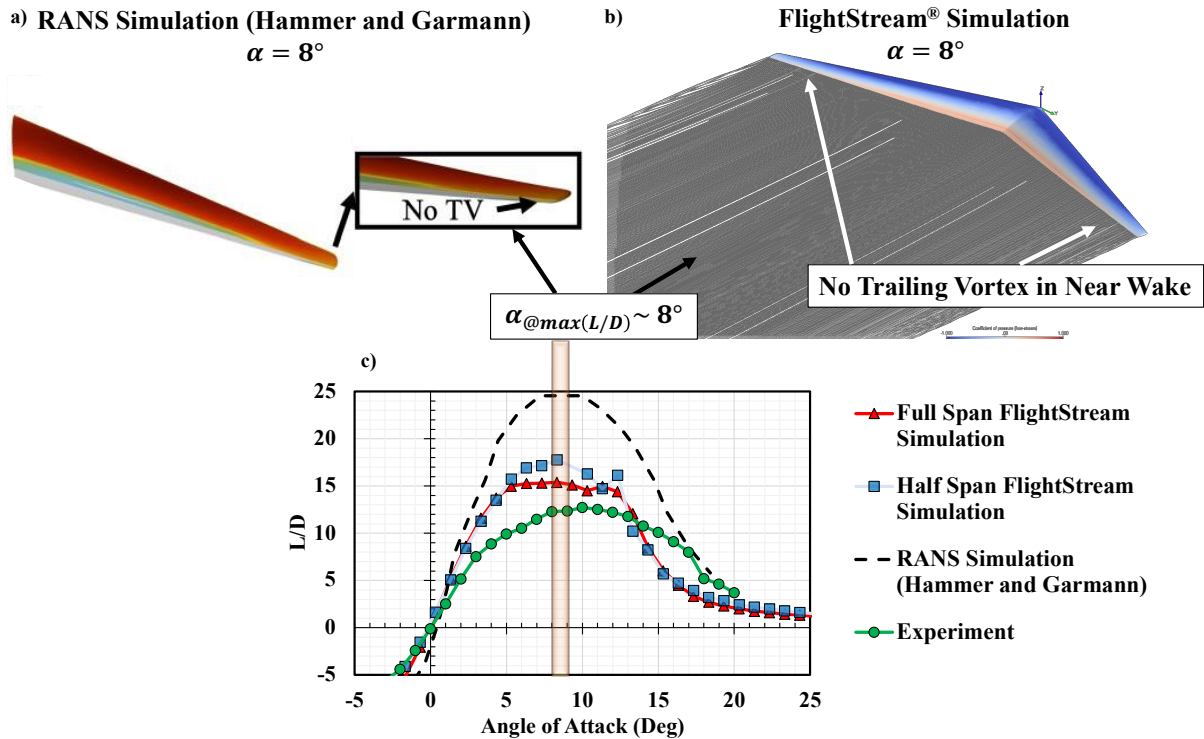
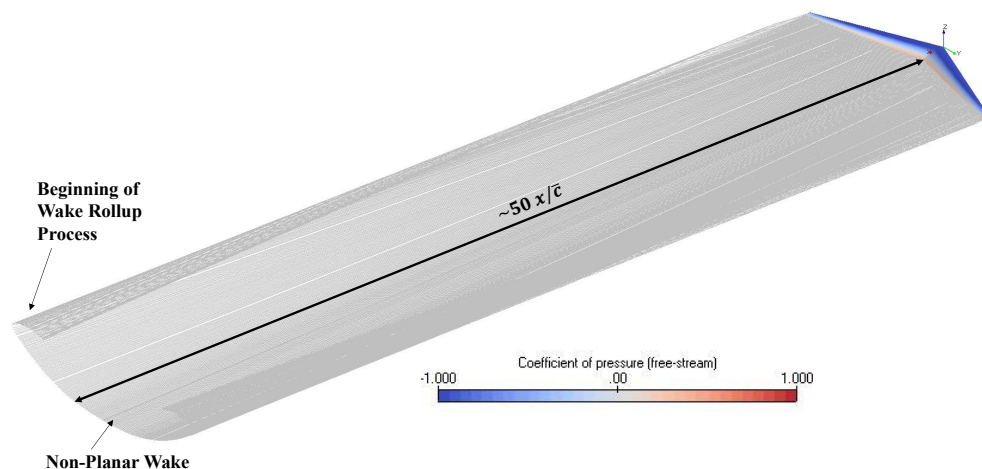


Figure 8 Simulation results a) from Hammer and Garmann [9] b) FlightStream® of PRANDTL-D3C wing at  $8^\circ$  angle of attack. c) Variation of aerodynamic efficiency as a function of angle of attack from FlightStream®, experiments, and RANS simulation from Hammer and Garmann [9]

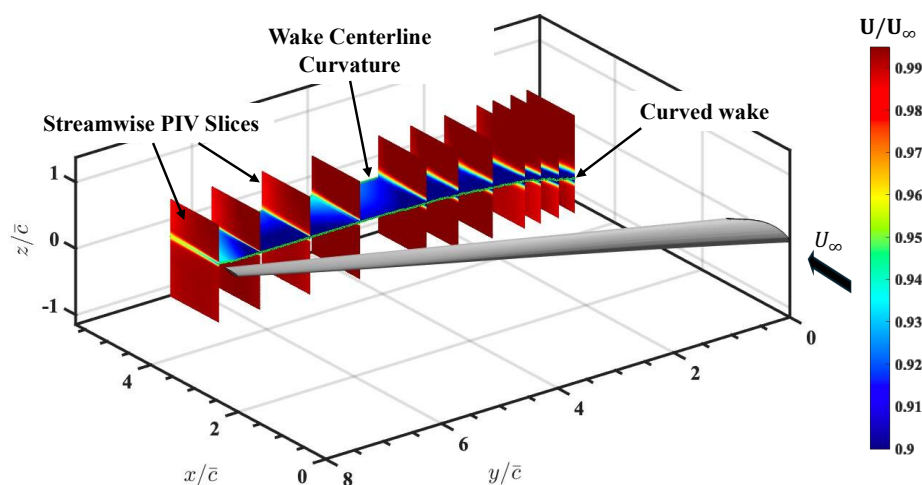


**Figure 9 Schematic of FlightStream® simulation at 8° angle of attack showing non-planar wake and the delayed wake rollup process. This result is in good agreement with results from Figure 4.**

The delayed wake rollup process observed by Ranjan and Ansell [8] can also be replicated by FlightStream®, which operates based on the panel method. The delayed wake rollup is observed in the wake of the Prandtl-D at an 8° angle of attack, as shown in Figure 9. According to the FlightStream® simulations, one full rotation of the trailing vortex occurs at approximately  $100 x/c$ , which aligns well with the results published by Ranjan and Ansell [8].

## B. PIV Results

The streamwise PIV data obtained at different spanwise stations, shown in Figure 10, were used to create a volume of data within the region of acquisition. A linear interpolation was performed along the wingspan to fill in the gaps between the PIV slices. Due to the significant non-linear changes in local angle of attack across the span and the different wake age due to sweep (downwash will carry the wake lower further downstream), the wake is inherently curved and twisted. To compare the momentum deficit, coefficient of drag, and the  $U$  and  $W$  velocities in the wing wake at different angles of attack, a planar slice across the wingspan is not ideal, as the magnitude of upwash and downwash of the wake change with  $\alpha_{wing}$ . Therefore, the wake centerline curvature was identified by tracking the minimum velocity in the wake across the downstream distance ( $x/\bar{c}$ ) to compare these quantities as a function of angle of attack.



**Figure 10 Wake velocity contours across different spanwise stations**



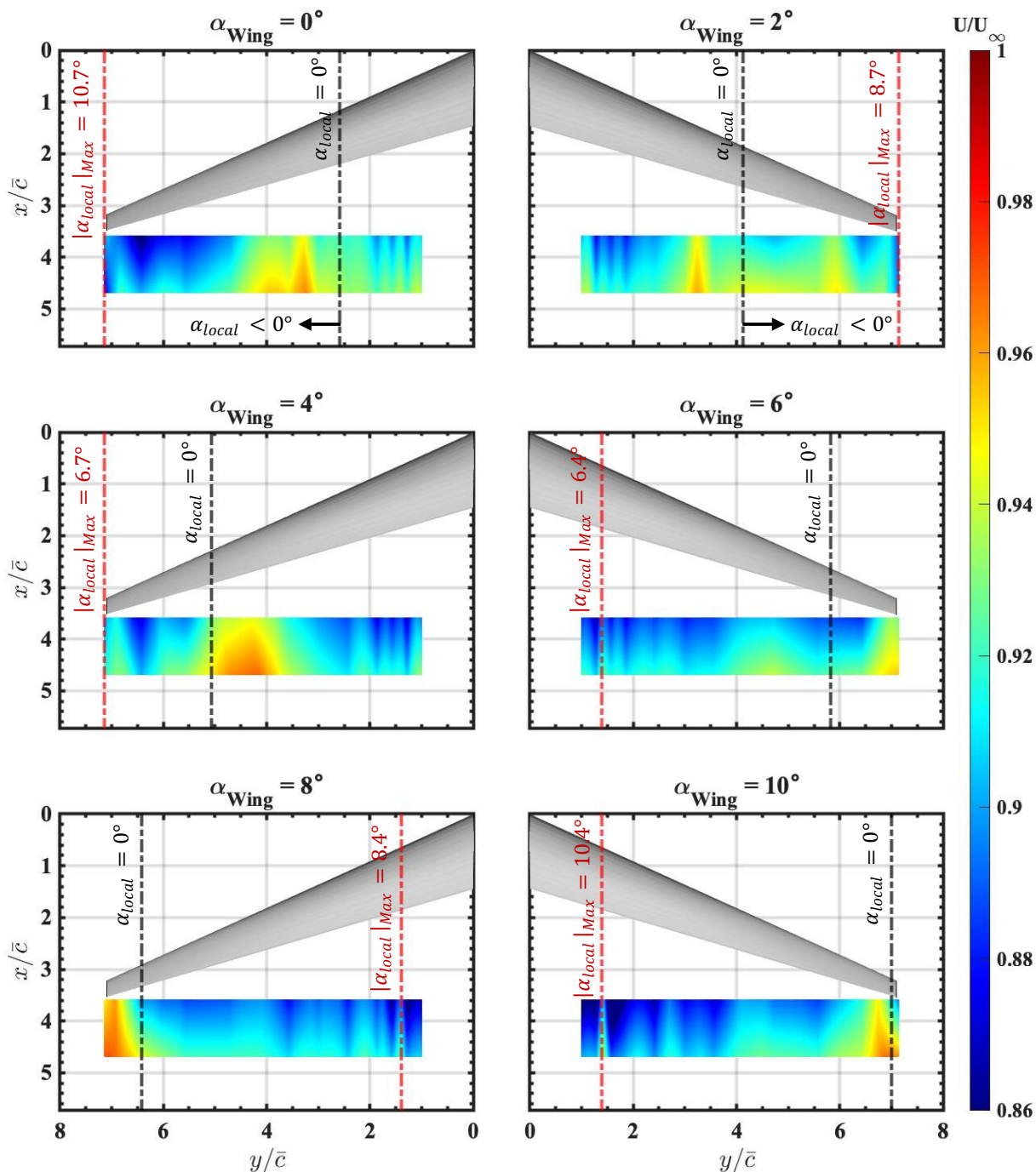
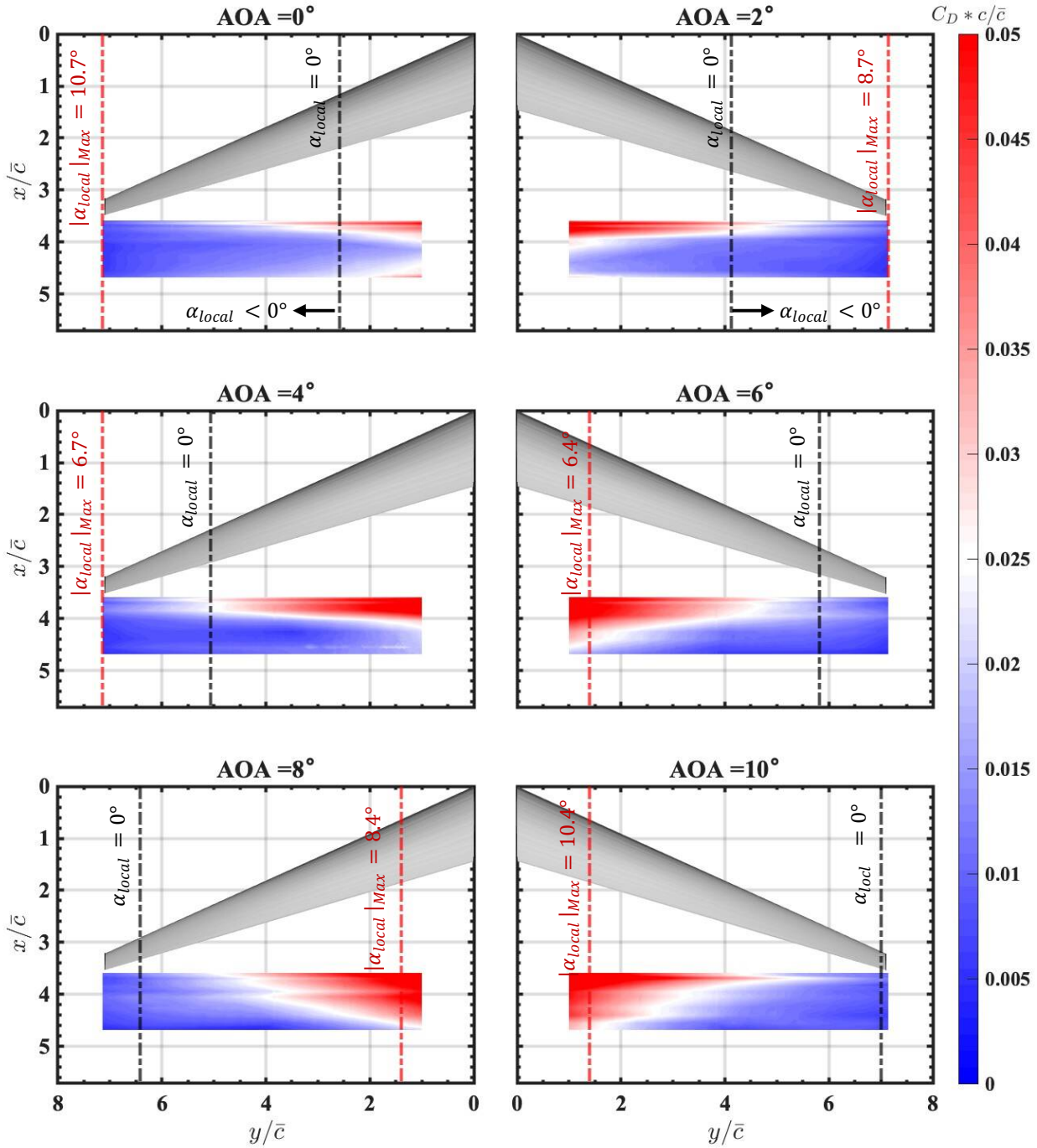


Figure 11 Variation of wake velocity on the wake centerline curvature along the wingspan at different  $\alpha_{Wing}$

The top view of the wake velocity along the wake centerline curvature is shown in Figure 11 for all angles of attack tested. The variation in minimum velocity at the wake centerline shows significant differences in the spanwise direction, which could be attributed to the higher degree of wing twist on the PRANDTL-D3C model. At  $\alpha_{Wing} = 0^\circ$ , the wing root is at a  $0^\circ$  angle of attack, while the wingtip is at  $-10^\circ$ . In Figure 11, the spanwise station where the local angle of attack  $\alpha_{local}$  reaches  $0^\circ$  is illustrated by the black dashed line. As  $\alpha_{Wing}$  increases, the  $\alpha_{local} = 0^\circ$  moves towards the wingtip, as expected. Coincidentally, a higher wake velocity (indicating a minimum momentum

deficit) is observed near the  $\alpha_{local} = 0^\circ$  station. The location of minimum momentum deficit follows the spanwise location of  $\alpha_{local} = 0^\circ$ .

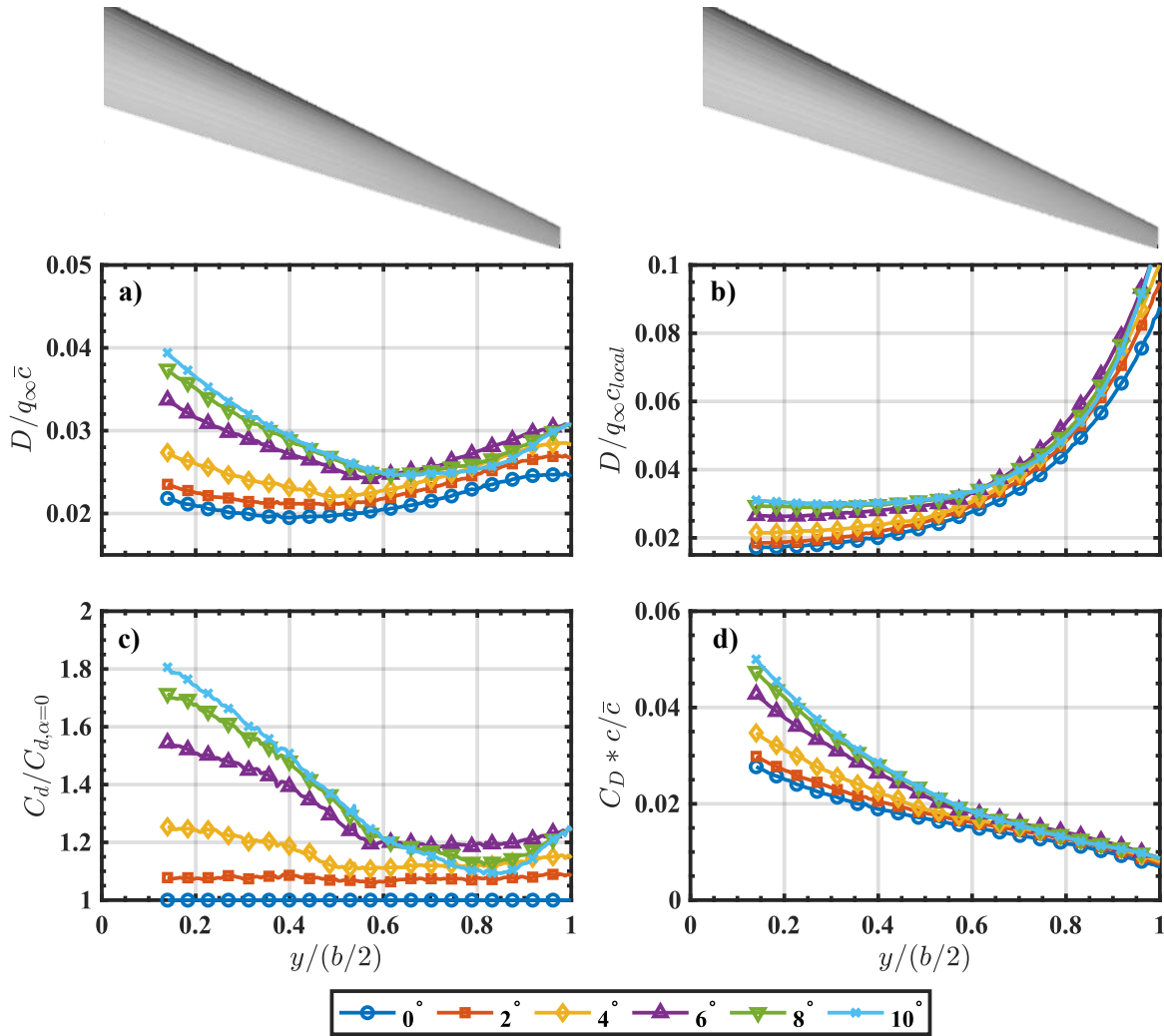


**Figure 12** Variation of coefficient of drag determined by momentum deficit on the wake centerline curvature along the wingspan at different  $\alpha_{Wing}$

The spanwise location where the absolute maximum local angle of attack occurs is illustrated by the red dashed line in Figure 11. For  $\alpha_{Wing} = 0^\circ, 2^\circ$ , and  $4^\circ$ , the absolute maximum angle of attack occurs at the wingtip as induced by the nonlinear twist distribution, which may explain why a higher momentum deficit (lower wake velocity) is

observed closer to the wingtip rather than the wing root. However, at angles of attack greater than  $4^\circ$ , the absolute maximum angle of attack is observed at the wing root.

To investigate the drag performance of the wing, Figure 12 shows contours of  $c/\bar{c}$  normalized momentum deficit determined drag coefficient along the wake centerline curvature as a function of wingspan at different angles of attack. The relaxation of the wake as a function of downstream distance is clearly visible at  $\alpha_{wing} = 0^\circ, 2^\circ$ , and  $4^\circ$ , especially near the wing root. The contours also show a spanwise gradient, with a higher normalized drag coefficient occurring at the wing root. However, the location of the maximum drag coefficient does not always correspond to the location of the maximum local absolute angle of attack. Furthermore, the normalized drag coefficient approaches zero toward the wingtips. Similar observations were made by Hammer and Garmann [9]. However, at  $8^\circ \alpha_{wing}$ , the RANS results showed a slight  $c/\bar{c}$  negative normalized drag coefficient, which is not observed in the current study.



**Figure 13** a) Drag coefficient along the wingspan normalized by a) MAC, b) local chord, c) Drag coefficient normalized by drag coefficient at  $\alpha = 0$  at different  $\alpha_{wing}$ , and d)  $\bar{c}/c$ .

A streamwise average of the drag coefficient was determined along the span to analyze the changes with angle of attack using different normalizations, as shown in Figure 13. The differences in normalization techniques highlight the sensitivity of drag distribution to chord length variations. Figure 13a shows the drag normalized by  $\bar{c}$ , and Figure 13b shows the drag coefficient normalized by the local chord. Normalizing by the local chord (Figure 13b) appears less effective in capturing momentum deficit changes due to the varying chord lengths across the span, while the

global chord normalization in Figure 13a better reflects drag trends. However, all cases display a nonlinear drag distribution, mirroring the downwash characteristics associated with the bell-shaped lift distribution. This finding is consistent with Hammer and Garmann's [9] study and reflects the downwash distribution associated with the bell-shaped lift distribution.

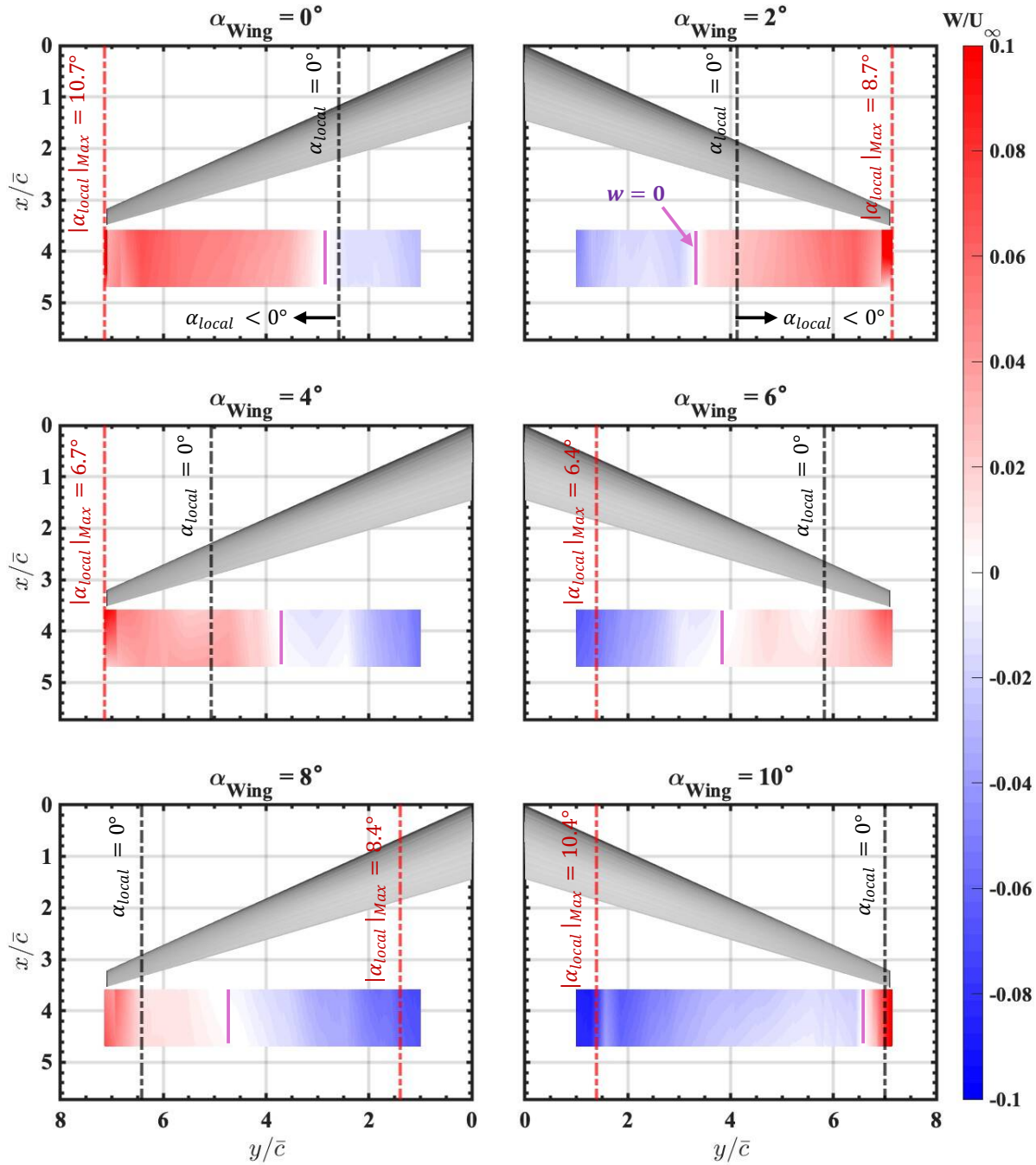


Figure 14 Variation of  $W/U_\infty$  velocity on the wake centerline curvature along the wingspan at different  $\alpha_{Wing}$

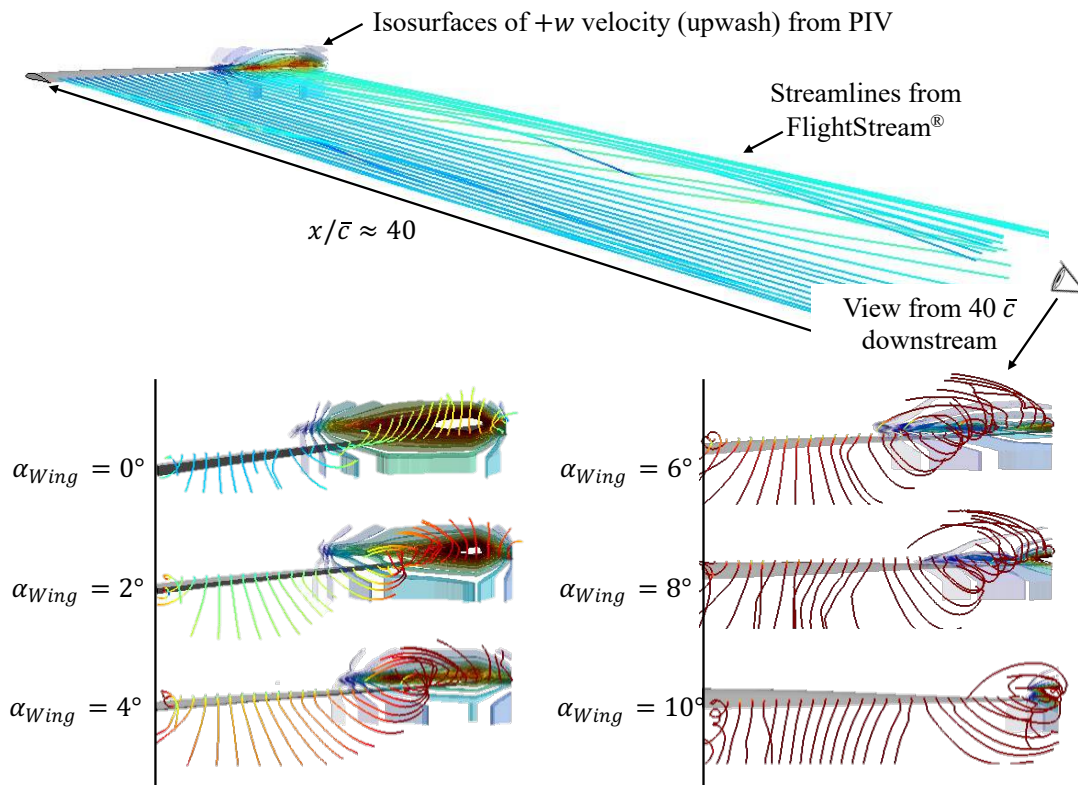
At  $\alpha_{Wing} = 0^\circ$ , the location of minimum drag is positioned at  $y/(b/2) = 0.4$ . Increasing the angle of attack shifts the location of minimum drag further toward the wingtip, which could be correlated to the movement of the wing vortex. As  $\alpha_{Wing}$  increases, the drag coefficient also rises at all span locations, except at  $8^\circ$  and  $10^\circ$ , where a reduction in drag starting from 60% of the span is observed. This trend is particularly evident in Figure 13c, which shows the



drag coefficient normalized by the  $0^\circ$  case. Figure 13c also reveals that the increase in drag is uniform at  $2^\circ$  across all span locations; however, this increase becomes nonlinear after  $4^\circ$ , with a higher increment in drag observed closer to the root.

The normalized  $W$  component of velocity in the wake of the wing is shown in Figure 14 for each of the angles of attack tested. The same spanwise stations of  $\alpha_{Local} = 0^\circ$ , and  $|\alpha_{Local}|_{Max}$  shown in Figure 11 are included here, along with the spanwise location where  $W/U_\infty = 0$ , indicating the transition from downwash to upwash. It is hypothesized that under certain conditions a vortex forms at the spanwise location where  $W/U_\infty = 0$ . However, cross-stream PIV needs to be conducted to identify under what set of conditions are necessary for this to occur. Starting with the inboard segments of the wing, as the angle of attack increases, the blue region of downwash increases from a moderate magnitude within the first quarter span at  $0^\circ$  to a more intense magnitude at  $10^\circ$  that further extends along the span. The location and proximity of  $\alpha_{Local} = 0^\circ$  do not appear to impact the transition from downwash to upwash, as these locations become increasingly distant from each other between  $4^\circ$  and  $8^\circ$ , suggesting a strong three-dimensional velocity component responsible for their separation at these angles of attack.

The location of  $W/U_\infty = 0$  shifts gradually outboard from  $0^\circ$  to  $8^\circ$  and then makes a large jump from  $8^\circ$  to  $10^\circ$ , likely due to the nonlinear twist, which changes at a greater rate around  $y/\bar{c} = 0.7$ . As the upwash moves toward the outboard wing section, the magnitude of the vertical velocity increases to a maximum of  $W/U_\infty = 0.1$ , indicating very intense upwash at  $10^\circ$ . At  $8^\circ$ , approximately one-third of the wingspan produces upwash, while the remaining two-thirds generate downwash in the inboard segments. The tip region shows a strengthened area of upwash, but not as intense as at  $10^\circ$ , where there is a noticeable increase in upwash intensity near the tip.



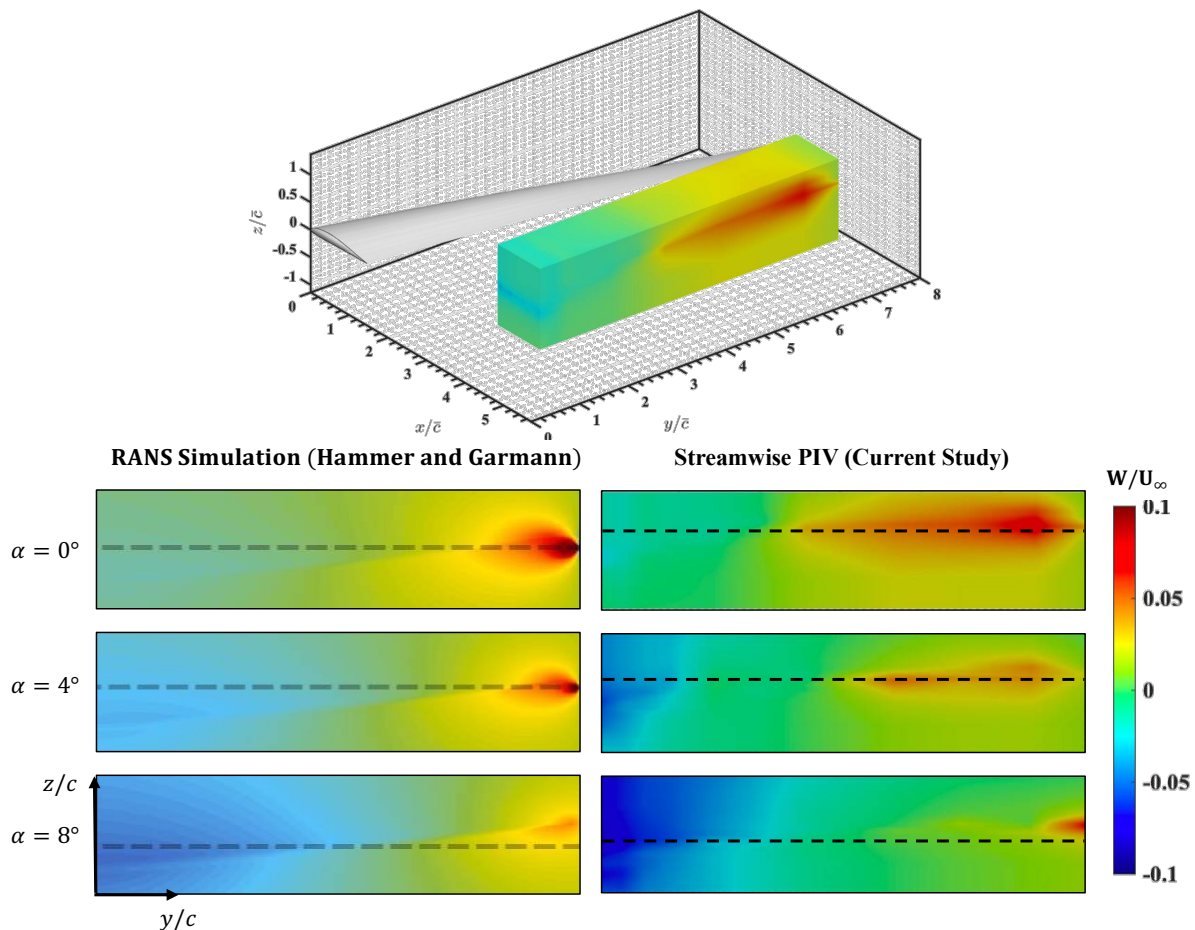
**Figure 15** Overlay of  $+W$  velocity contours from PIV and wake streamlines from FlightStream at different  $\alpha_{Wing}$

The PIV data was used to generate isosurfaces of the  $W$  velocity, while the FlightStream<sup>®</sup> data was used to create long-form streamlines in the wing's wake, extending up to 40 chord lengths downstream. Together, in Figure 15, this overlay of data shows a relationship between the near-wake velocity measurements and the far-wake vortex rollup at  $\sim 40 x/\bar{c}$  if such a superposition can be entertained. The alignment between the PIV isosurfaces and the spanwise



location of upwash growth with the downstream wake rollup of the FlightStream<sup>®</sup> streamlines is in good agreement. It should be noted that the rear view of the wake streamlines represented in Figure 15 is more than 40 chord lengths away and for most angles of attack, one full rollup of the vortex is not complete. Nevertheless, the formation of upwash appears to play a critical role in the inboard movement of the wingtip vortex, indicating that the bell-shaped lift distribution effectively influences downstream vortex positioning. This connection between the wing twist and the formation of wing vortex is also seen in Schreyer et al. [17]. A dedicated cross-stream PIV study is needed to fully understand the extent to which the lift distribution, angle of attack, and associated upwash profiles play a role in the development and location of the wing vortex.

The W velocity data from the current study's PIV is compared to the Hammer and Garmann simulations [9] in Figure 16. There is strong agreement between the two with respect to the formation of an upwash region near the tip of the wing, which decreases as the angle of attack increases. Additionally, the growth of a downwash region near the inboard section of the wing is prominent in both data sets, with this region expanding as the angle of attack increases. Notably, the general trends in the global wake profile observed in the PIV data are captured by the RANS simulations, showing relatively similar regions of upwash and downwash behavior and magnitude across the span. This agreement builds confidence that the RANS simulations are accurately capturing trends in the wake flow field and can be used to further explore changes to the geometry and their impact on the wake profile.



**Figure 16 Comparison of normalized W velocity at  $x/\bar{c} = 3.2$  between RANS simulation ( $Re_{\bar{c}} = 3.43 \times 10^5$ ) [9] and PIV (current study)**

## IV. Conclusion

Force-based testing and near-wake streamwise PIV were conducted on a 1:11 scaled model of the PRANDTL-D3C wing at different angles of attack. The results were compared with FlightStream®, a panel-method-based simulation tool, and the RANS simulations by Hammer and Garmann [9]. The  $C_L$ ,  $C_D$ , and aerodynamic efficiency from the three methods showed good agreement, with differences in the stall angle attributed to differences in Reynolds numbers between the methods. All methods indicated that the maximum aerodynamic efficiency occurs around the design condition of the PRANDTL-D3C wing, at  $\alpha_{wing} = 8^\circ$ . At this angle of attack, the RANS simulations showed the absence of a tip vortex, which was also observed in the FlightStream® simulation. The simulations indicated the presence of wake rollup far downstream, beyond  $40 x/\bar{c}$ , and even then, the wake folded over rather than displaying conventional vortex rollup seen in the wakes of elliptically loaded wings.

The wake velocity along the wake centerline curvature showed a lower momentum deficit where the local angle of attack reached zero, which was consistent across all angles of attack, with lower momentum deficit observed near the wingtip. The  $c/\bar{c}$  normalized drag coefficient showed remarkable similarity across all angles of attack, with higher drag observed inboard while the wingtips remained lightly loaded and experienced less drag. This characteristic could be correlated to the maximum efficiency peak at  $\alpha_{wing} = 8^\circ$ . At this angle of attack, more than two-thirds of the wing produced downwash in the wake, while one-third generated upwash near the tip. The location of upwash in the near wake agreed well with the wake streamlines from the panel method confirming the predominantly inviscid nature of the wingtip vortex rollup process.

Although the streamwise PIV results were insufficient to confirm the tip vortex location found in the RANS simulations, the agreement between the simulation and experimental results provides greater confidence in the results of the RANS and FlightStream® simulations. A cross-stream study is currently underway to quantify x-vorticity in the wake of the PRANDTL-D3C wing.

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